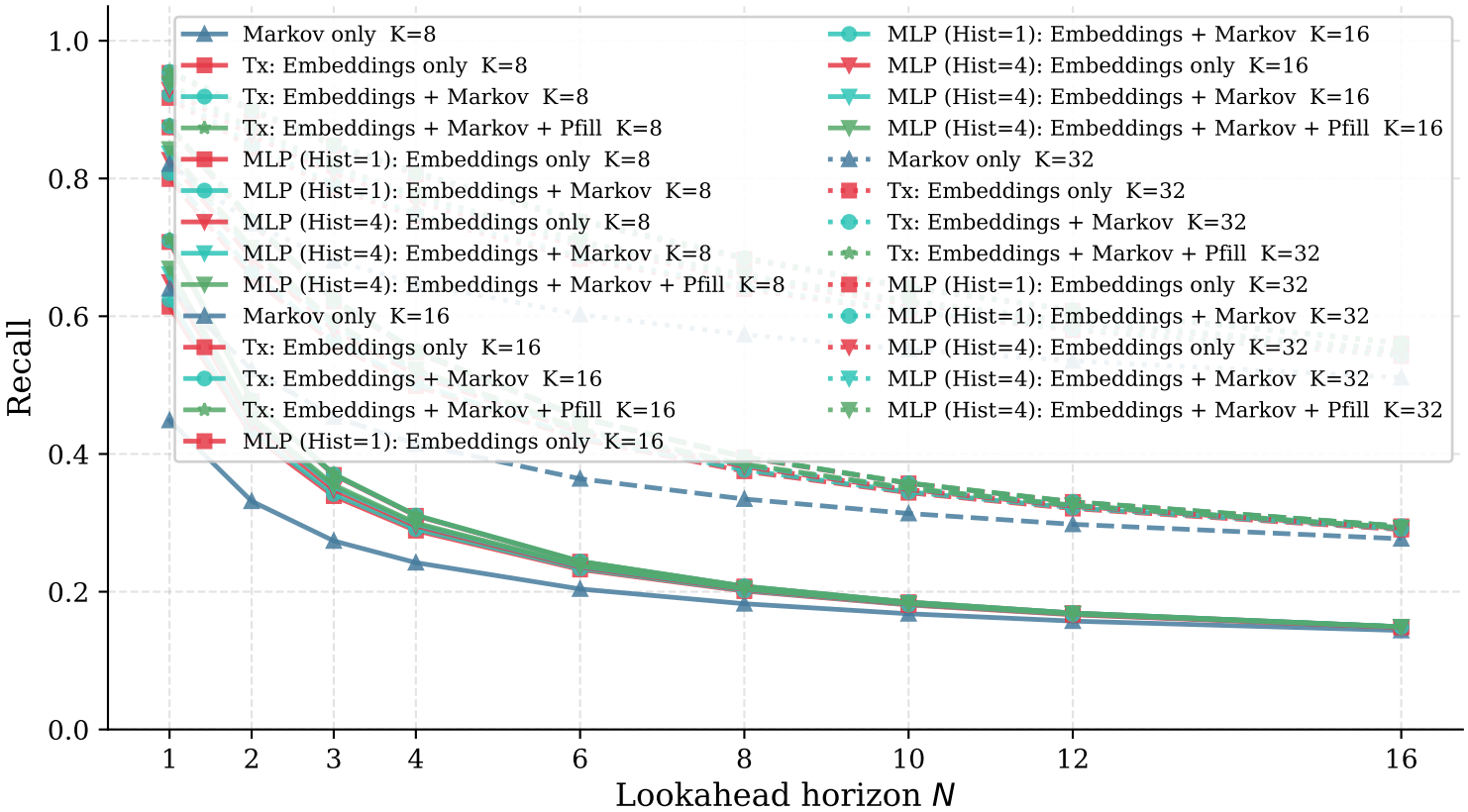

Fixed-K Recall vs Horizon — Ablation
(mean over layers, K ∈ [8, 16, 32])
Recall
Lookahead horizon N
0.0
0.2
0.4
0.6
0.8
1.0
1
2
3
4
6
8
10
12
16
Markov only K=8
Tx: Embeddings only K=8
Tx: Embeddings + Markov K=8
Tx: Embeddings + Markov + Pfill K=8
MLP (Hist=1): Embeddings only K=8
MLP (Hist=1): Embeddings + Markov K=8
MLP (Hist=4): Embeddings only K=8
MLP (Hist=4): Embeddings + Markov K=8
MLP (Hist=4): Embeddings + Markov + Pfill K=8
Markov only K=16
Tx: Embeddings only K=16
Tx: Embeddings + Markov K=16
Tx: Embeddings + Markov + Pfill K=16
MLP (Hist=1): Embeddings only K=16
MLP (Hist=1): Embeddings + Markov K=16
MLP (Hist=4): Embeddings only K=16
MLP (Hist=4): Embeddings + Markov K=16
MLP (Hist=4): Embeddings + Markov + Pfill K=16
Markov only K=32
Tx: Embeddings only K=32
Tx: Embeddings + Markov K=32
Tx: Embeddings + Markov + Pfill K=32
MLP (Hist=1): Embeddings only K=32
MLP (Hist=1): Embeddings + Markov K=32
MLP (Hist=4): Embeddings only K=32
MLP (Hist=4): Embeddings + Markov K=32
MLP (Hist=4): Embeddings + Markov + Pfill K=32